

James Buck

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Profile

Results-oriented Software Engineer with 12 years of experience in developing scalable web applications, RESTful APIs, and data-driven solutions. Proven expertise in **Python**, **Django**, **FastAPI**, **React.js**, and cloud-based microservices architecture. Adept at integrating NLP, data engineering, and ML pipelines into robust applications. Strong focus on security, performance, and scalable design, with experience across Agile environments.

Experience

SENIOR SOFTWARE ENGINEER | ELEMENT AI | MARCH 2022 - PRESENT

Element AI is an artificial intelligence company that gives organizations access to technology.

- Designed and developed AI-driven solutions using **FastAPI** and **Python**, enabling real-time data processing and seamless integration with client systems to address diverse business challenges.
- Collaborated with cross-functional teams to implement scalable architectures, ensuring reliability and performance across multiple AI-enabled applications.
- Enhanced deployment workflows by creating **CI/CD** pipelines with **Jenkins** and **Azure DevOps**, reducing integration time and minimizing deployment errors.
- Built and maintained scalable cloud solutions using **AWS** services like **EC2**, **S3**, and **Lambda**, achieving high availability and cost efficiency.
- Optimized machine learning workflows by integrating tools like **Apache Spark** and **Databricks** for faster data analysis, reducing processing times by 40%.
- Delivered technical training sessions for client teams, enabling them to adopt and utilize AI-based solutions effectively, improving system adoption rates.
- Wrote comprehensive technical documentation for APIs and deployment workflows, ensuring ease of maintenance and knowledge transfer for internal and client teams.
- Worked with senior leadership to identify and implement solutions aligned with client business goals, delivering measurable value through AI and automation.

SENIOR SOFTWARE ENGINEER (L5) | GOOGLE | JUNE 2018 – MARCH 2022

- Developed and maintained RESTful APIs for Google Cloud Platform, enabling seamless data integrations and empowering millions of users with real-time cloud-based solutions.
- Designed high-performance data pipelines for large-scale analytics, leveraging **BigQuery** and **Cloud Storage**, which improved data retrieval efficiency by over 30%.
- Built responsive and reusable front-end components using **React.js**, ensuring consistency across interfaces and delivering a smooth user experience for enterprise clients.
- Spearheaded the migration of monolithic applications to **Kubernetes**-based microservices, improving scalability and reducing deployment time for critical workloads.
- Implemented robust security measures using **OAuth2** and role-based access control (**RBAC**), ensuring secure communication between services and compliance with industry standards.

- Conducted thorough code reviews and collaborated with cross-functional teams to establish best practices for maintaining code quality and improving team productivity.
- Partnered with product managers and designers to refine requirements, deliver features aligned with business objectives, and improve customer satisfaction.
- Provided detailed performance optimization for backend systems, achieving a significant reduction in latency for high-traffic applications.

FULL STACK DEVELOPER | CLARIFAI | OCTOBER 2015 – MAY 2018

A leading, full-lifecycle deep learning AI platform for computer vision, natural language processing, and audio recognition.

- Built and maintained machine learning-backed web applications using **Python**, **Django**, and **React.js**, focusing on scalability and high performance for AI-based solutions.
- Designed user-centric front-end interfaces, integrating **Redux** and **React Hooks** for efficient state management, enhancing the responsiveness and usability of applications.
- Developed RESTful APIs that facilitated the integration of Clarifai's computer vision models into third-party applications, increasing platform adoption by enterprise clients.
- Enhanced database performance by optimizing complex queries and indexes in **PostgreSQL**, improving response times for large-scale datasets.
- Conducted unit and integration testing using **Pytest**, ensuring code reliability and reducing deployment issues in production.
- Partnered with data scientists to integrate trained models into production environments, enabling real-time image and video recognition capabilities.
- Led technical discussions with stakeholders to translate business requirements into technical deliverables, ensuring alignment with client expectations.
- Provided ongoing maintenance and support for production systems, proactively addressing issues to ensure uninterrupted service availability.

SOFTWARE ENGINEER | HEALTH CATALYST | MARCH 2012 – OCTOBER 2015

A tech platform that organizes and links health-related data from different systems.

- Designed and implemented backend services for healthcare data platforms using **Flask** and **Python**, enabling secure and efficient management of patient information and claims processing.
- Built and optimized ETL pipelines for ingesting, transforming, and storing large datasets in compliance with healthcare data standards, improving processing efficiency by 25%.
- Developed interactive dashboards with **Matplotlib** and **Plotly**, allowing healthcare providers to visualize patient trends and improve care delivery.
- Integrated secure authentication mechanisms with **OAuth2** and **JWT**, ensuring HIPAA-compliant data access and maintaining privacy standards.
- Conducted performance tuning for **PostgreSQL** databases, optimizing complex queries and improving data retrieval speed for high-traffic systems.
- Automated report generation and monitoring workflows, reducing manual intervention and improving operational efficiency for healthcare clients.
- Collaborated with cross-functional teams to gather requirements, create technical specifications, and deliver user-friendly features within Agile sprints.

- Created comprehensive system documentation to facilitate knowledge sharing and ensure long-term maintainability of healthcare solutions.

Education

BACHELOR OF SCIENCE IN COMPUTER SCIENCE | APRIL 2008 – MARCH 2012 | TOWSON UNIVERSITY

Skills & Abilities

- Languages: Python, JavaScript (React.js, Node.js), SQL, TypeScript, Go, Bash
- Frameworks and Libraries: FastAPI, Flask, Django, React.js, Redux, Pytest, NumPy, Pandas, TensorFlow, Scikit-learn
- Databases: PostgreSQL, MySQL, MongoDB, BigQuery, Redis, Elasticsearch
- Cloud and DevOps: AWS (EC2, S3, Lambda, RDS, DynamoDB), Google Cloud Platform (BigQuery, Cloud Functions), Azure, Jenkins, Kubernetes, Docker, Terraform
- Tools and Platforms: Databricks, Apache Spark, Splunk, Matplotlib, Plotly, Airflow, Apache Kafka
- Development Practices: Agile (Scrum), Test-Driven Development (TDD), Continuous Integration/Continuous Deployment (CI/CD), Version Control (Git)
- Soft Skills: Team Leadership, Cross-functional Collaboration, Technical Documentation, Requirements Gathering, Stakeholder Communication